

TightLLM: Maximizing Throughput for LLM Inference via Adaptive Offloading Policy

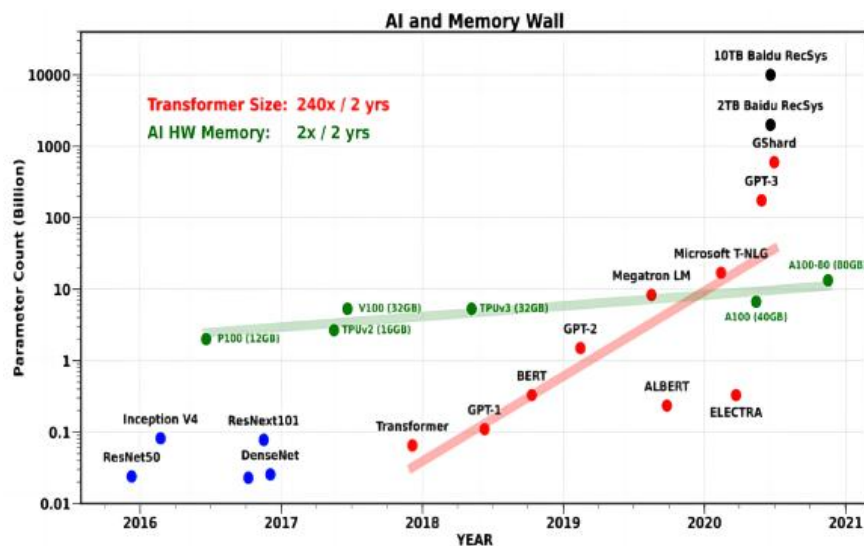
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Zhixin Zhao, Sheng Chen, Laiping Zhao, Wenxin Li, Keqiu Li

IEEE Transactions on Computers

挑战：大模型推理服务

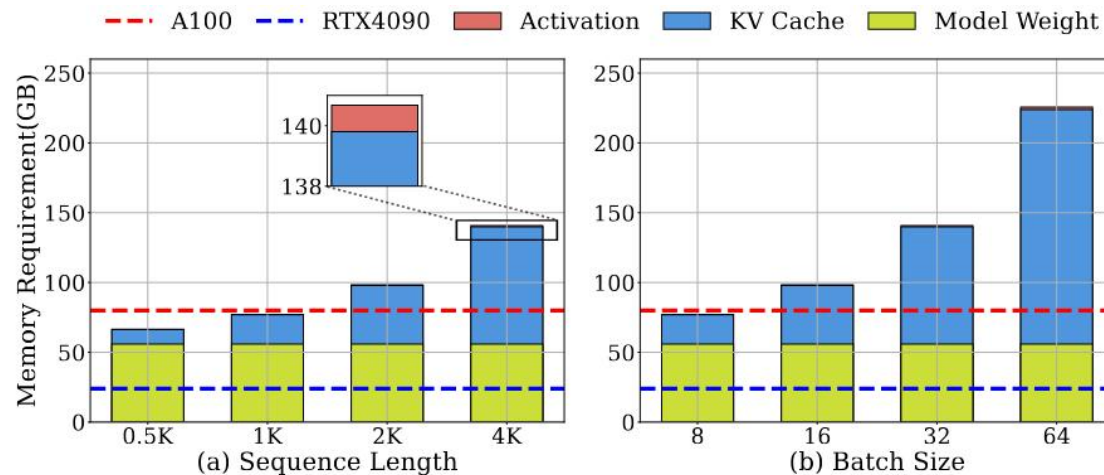
单个GPU显存不足以支持大型模型的推理需求

芯片提升无法适应需求增长



模型规模增速**240x/2yrs**，硬件显存增速**2x/2yrs**

现有单个GPU无法部署大模型

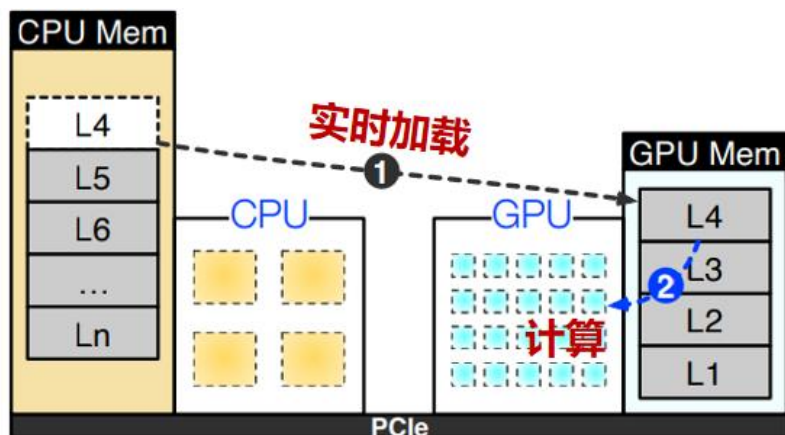


OPT-30B所需的显存远超**消费级/服务器级GPU**

现有方法：LLM卸载

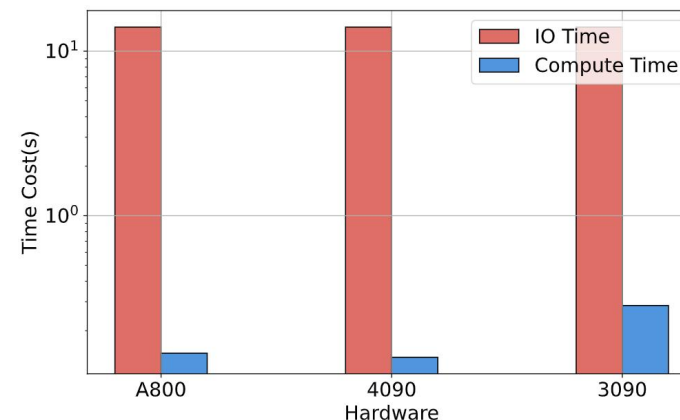
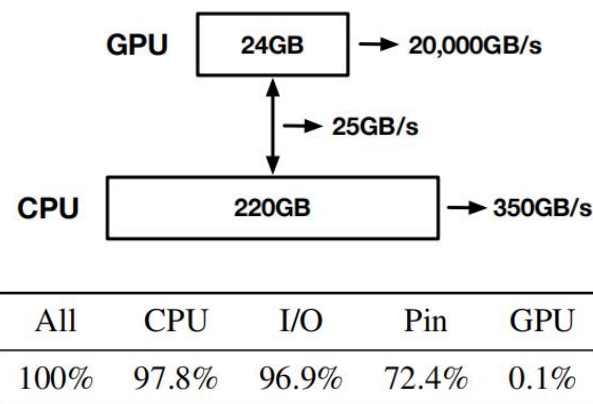
通过卸载技术降低显存需求，然而会带来额外的I/O开销

卸载至低级存储^{[1][2]}



- 张量卸载到**CPU内存及硬盘**
- 轮流加载**所需张量完成推理

额外开销：CPU到GPU的数据传输



- CPU和GPU间通信速率慢，I/O在整个推理过程中占比高**
- GPU计算需要等待张量传输，导致计算资源严重浪费**

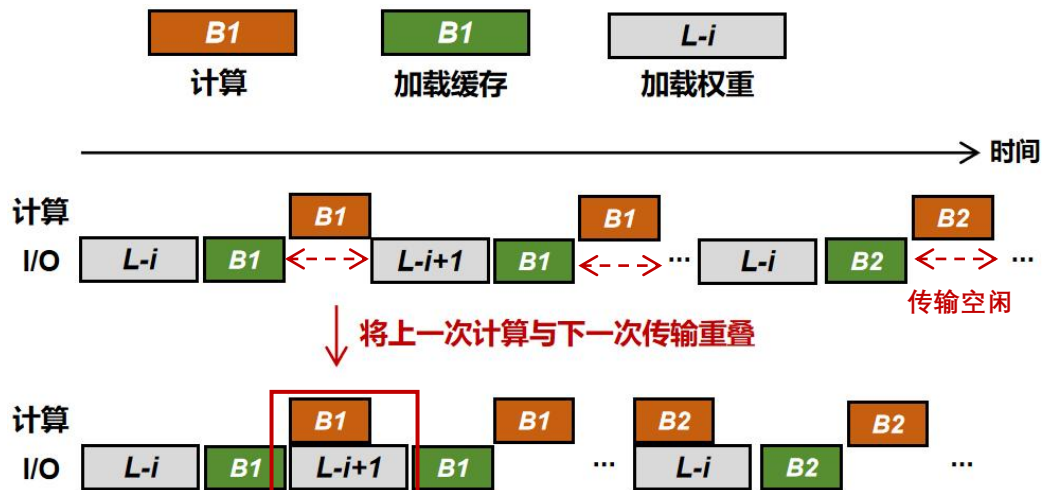
[1] Song Y, Mi Z, Xie H, et al. Powerinfer: Fast large language model serving with a consumer-grade gpu[J]. arXiv preprint arXiv:2312.12456, 2023.

[2] Lee W, Lee J, Seo J, et al. {InfiniGen}: Efficient generative inference of large language models with dynamic {KV} cache management[C]//18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24). 2024: 155-172.

现有优化工作^[1]

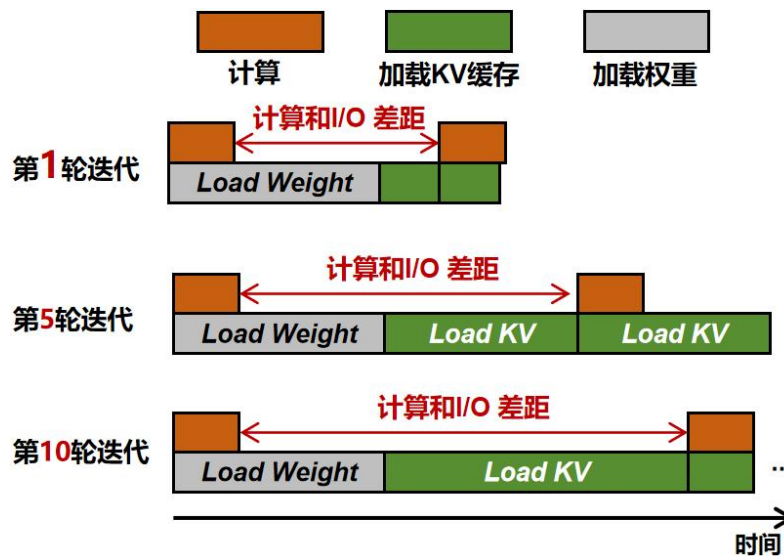
实现计算与数据传输的重叠

计算和传输重叠



- 计算和传输重叠：预加载权重和缓存
- 降低传输开销：将传输隐藏在计算中

现有工作重叠效果差



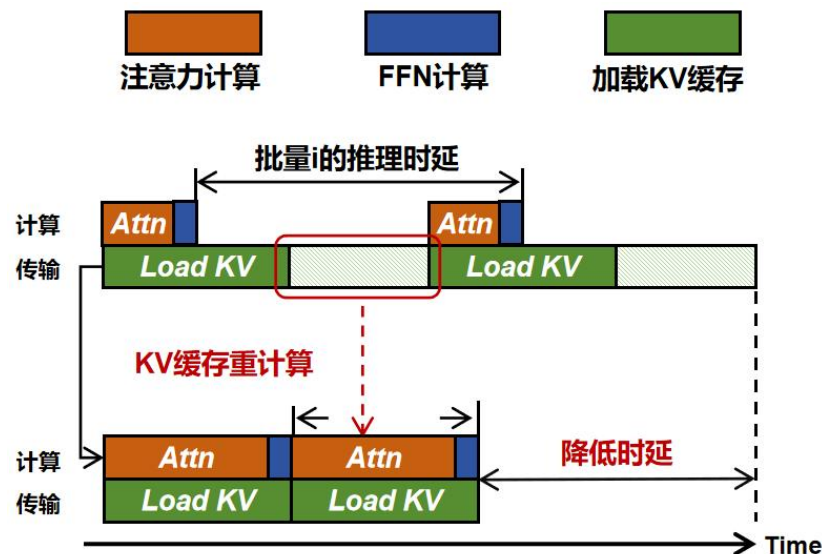
- KV缓存带来不断增长的计算/传输**差距**
- 权重加载**加剧**了传输带来的I/O瓶颈

[1] Sheng Y, Zheng L, Yuan B, et al. Flexgen: High-throughput generative inference of large language models with a single gpu[C]//International Conference on Machine Learning. PMLR, 2023: 31094-31116.

解决方案：以算换传结合流水线加载

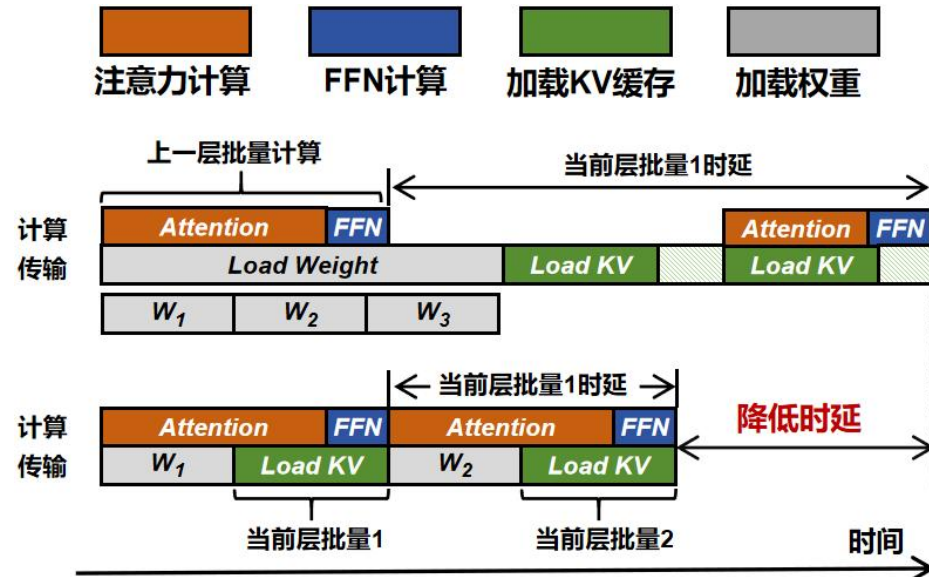
通过对KV缓存进行重计算，实现计算替代数据传输，并采用流水线方式加载权重

重计算KV缓存



- 重计算部分KV缓存代替传输缓存开销
- 调整重计算比例实现最佳计算-传输重叠

流水线加载权重

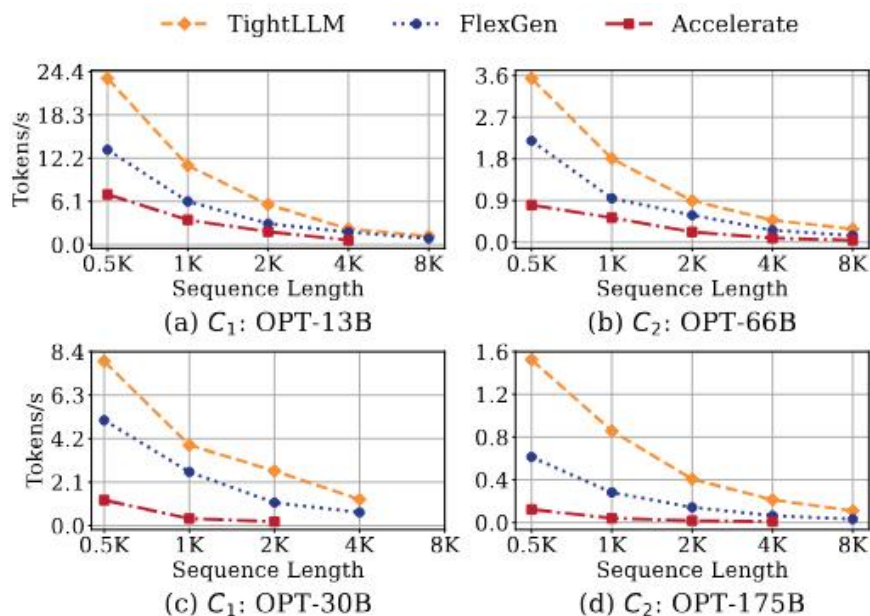


- 流水线加载权重利用多个批量摊销开销
- 结合缓存重计算进一步降低时延

整体方案评估

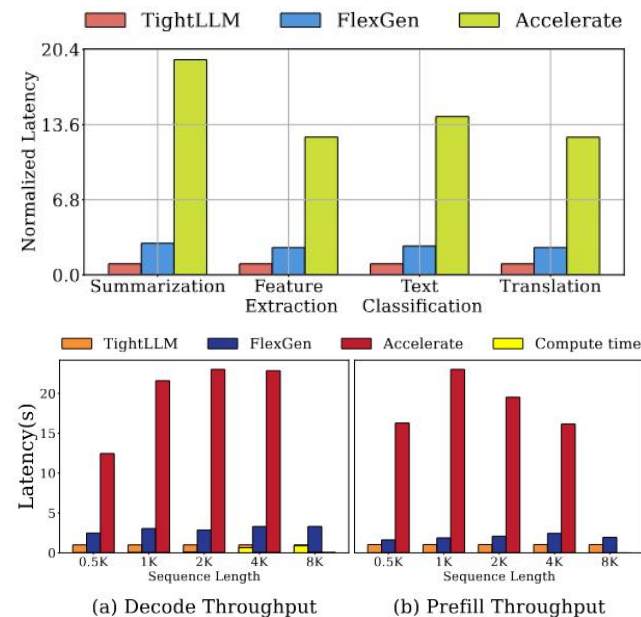
实验评估——Token处理/生成吞吐量，任务完成时间

高吞吐



- 吞吐量比现有推理系统提高**1.3x-22x**

大规模任务完成时间



- 大规模任务完成时间降低**59.6%-94.9%**
- 推理过程中传输时间降低**19.9%-97.5%**